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LAB ASSESSMENT 8

Agile Development Process and Devops Lab

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Course code:ISWE406P

Lab slot:L45+L46

Kubernetes Integration in a CI/CD Pipeline

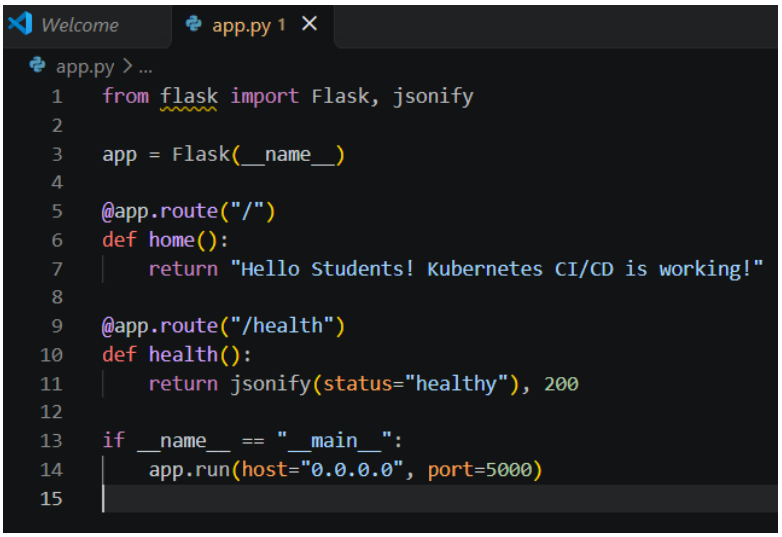
Create the sample Flask application:

Step 1-Create the project folder:

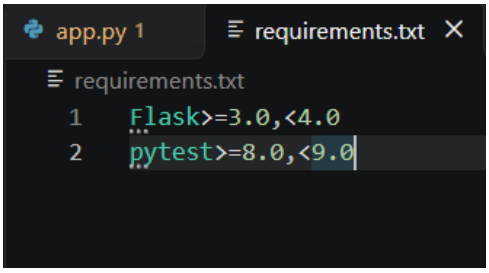
Create a folder named student-k8s-demo and open it in VS Code. Create the following structure:

```
student-k8s-demo/
■■■ app.py
■■■ requirements.txt
■■■ test_app.py
■■■ Dockerfile
■■■ .dockerignore
■■■ .gitignore
■■■ k8s/
■    ■■■ deployment.yaml
■    ■■■ service.yaml
■■■ .github/
    ■■■ workflows/
        ■■■ pipeline.yml
```

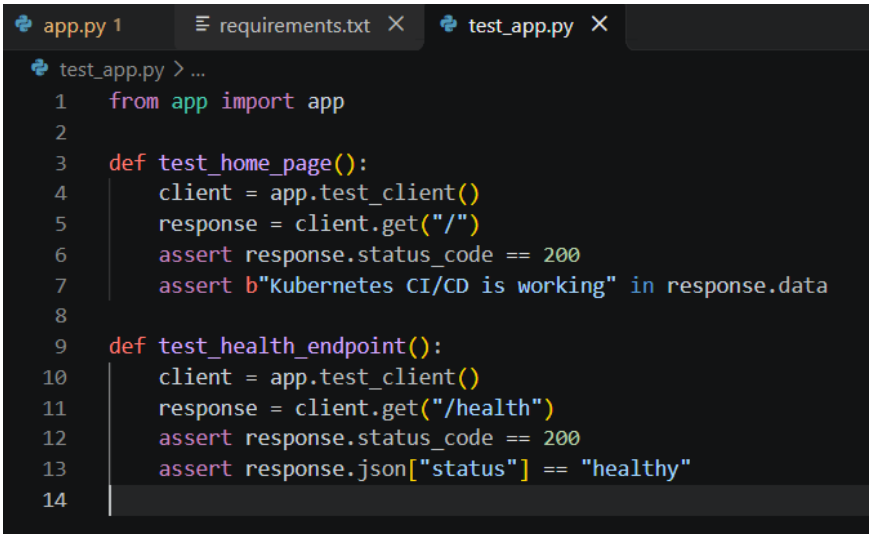
Step 2 -Create app.py



Step 3- Create requirements.txt



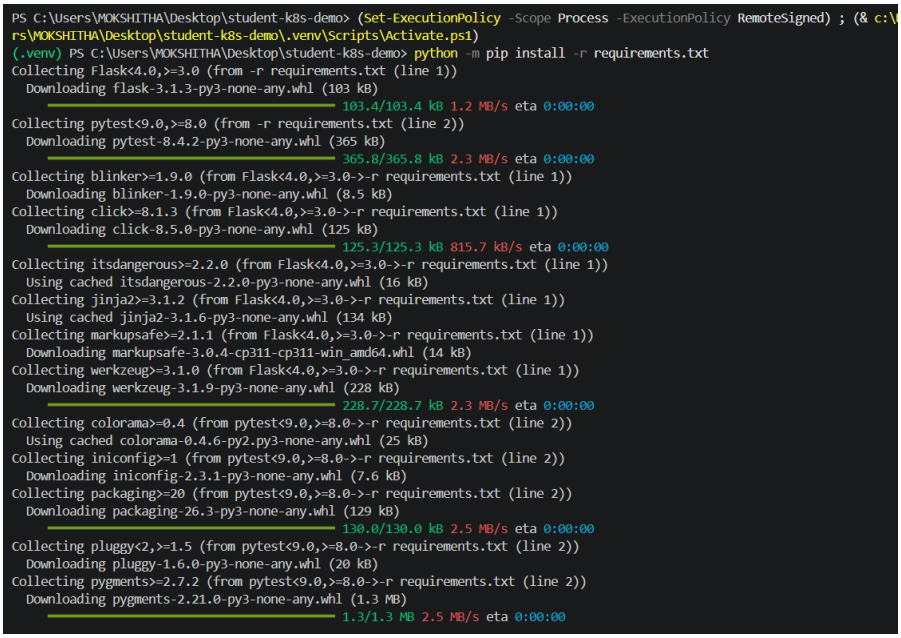
Step 4 -Create test_app.py



Step 5 - Run locally

1.Open Terminal->New Terminal and give the command as “python -m venv .venv”

2. Open another New Terminal and give commands like this:

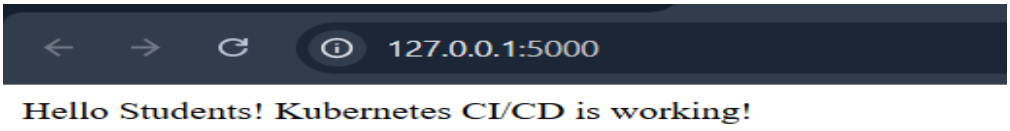


```
Installing collected packages: pygments, pluggy, packaging, markupsafe, itsdangerous, iniconfig, colorama, click, blinker, wer
kzeug, pytest, jinja2, Flask
Successfully installed Flask-3.1.3 blinker-1.9.0 click-8.5.0 colorama-0.4.6 iniconfig-2.3.1 itsdangerous-2.2.0 jinja2-3.1.6 ma
rkupsafe-3.0.4 packaging-26.3 pluggy-1.6.0 pygments-2.21.0 pytest-8.4.2 werkzeug-3.1.9

[notice] A new release of pip is available: 23.1.2 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> ^C
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> python -m pytest -v
============================================= test session starts =============================================
platform win32 -- Python 3.11.4, pytest-8.4.2, pluggy-1.6.0 -- C:\Users\MOKSHITHA\Desktop\student-k8s-demo\.venv\Scripts\pytho
n.exe
cachedir: .pytest_cache
rootdir: C:\Users\MOKSHITHA\Desktop\student-k8s-demo
collected 2 items

test_app.py::test_home_page PASSED                                                                    [ 50%]
test_app.py::test_health_endpoint PASSED                                                              [100%]

============================================== 2 passed in 0.27s ==============================================
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> python app.py
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.16.156.190:5000
Press CTRL+C to quit
127.0.0.1 - - [14/Oct/2026 18:49:34] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [14/Oct/2026 18:49:34] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [14/Oct/2026 18:49:53] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [14/Oct/2026 18:49:53] "GET /favicon.ico HTTP/1.1" 404 -
```



Build and run the Docker image:

Step 6 -Create Dockerfile

app.py 1 | 127.0.0.1:5000 | Dockerfile | requirements

Dockerfile

```
1   FROM python:3.12-slim
2
3   WORKDIR /app
4
5   COPY requirements.txt .
6   RUN pip install --no-cache-dir -r requirements.txt
7
8   COPY app.py .
9
10  EXPOSE 5000
11
12  CMD ["python", "app.py"]
13
```

Step 7 -Create .dockerignore

.dockerignore

```
.venv/
__pycache__/
.pytest_cache/
.git/
```

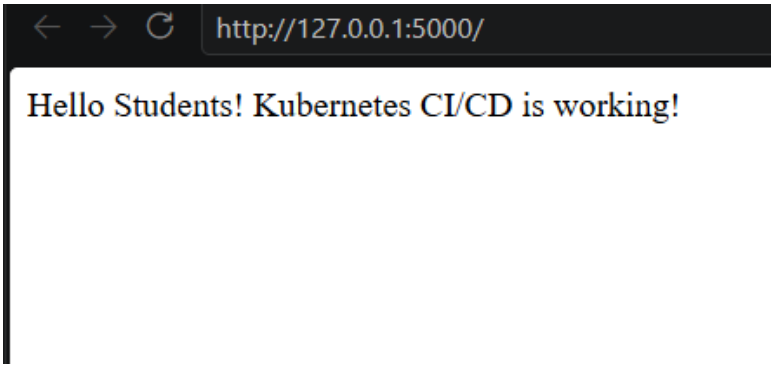
Step 8 -Build the image

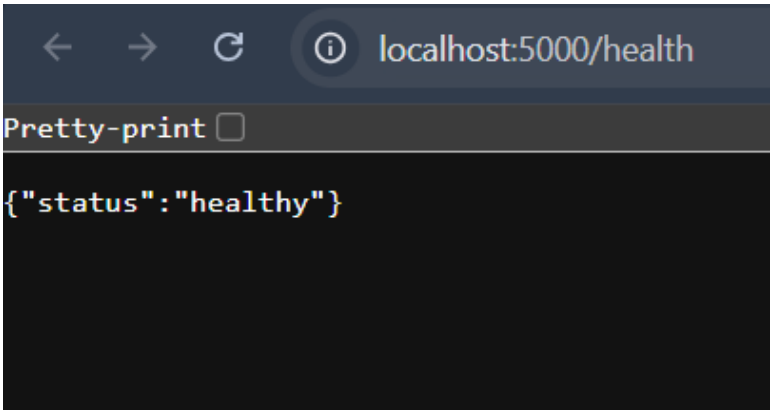
```
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> docker build -t student-k8s-demo:v1 .
[+] Building 48.9s (11/11) FINISHED                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                  0.0s
 => => transferring dockerfile: 217B                                                                                  0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                                   5.7s
 => [auth] library/python:pull token for registry-1.docker.io                                                         0.0s
 => [internal] load .dockerignore                                                                                     0.0s
 => => transferring context: 85B                                                                                      0.0s
 => [1/5] FROM docker.io/library/python:3.12-slim@sha256:a6e34c598f2467ed0e9a8d349809fcd8b5c603269512df273a0bb1784edc1  20.2s
 => => resolve docker.io/library/python:3.12-slim@sha256:a6e34c598f2467ed0e9a8d349809fcd8b5c603269512df273a0bb1784edc11  0.0s
 => => sha256:bbc5a12237474ec0b23b5051260d56d6436a24d70d9cad961adebe7a037ed813 249B / 249B                            0.4s
 => => sha256:6912e23eb44595b44dc6d9882b34f305ad6f6e9667cec70b13e5dbc4f8cd97f4 12.12MB / 12.12MB                     11.3s
 => => sha256:75b6a36c64a1cf3520fc4b4aae4739d5caa8a6a05c27835a920a61784c9c32f8 1.29MB / 1.29MB                        2.4s
 => => sha256:ecc510c1e359bdc007b39802570bb7e4dec7d6ebccf357c3201a74299113e05e 29.84MB / 29.84MB                     18.3s
 => => extracting sha256:ecc510c1e359bdc007b39802570bb7e4dec7d6ebccf357c3201a74299113e05e                             1.1s
 => => extracting sha256:75b6a36c64a1cf3520fc4b4aae4739d5caa8a6a05c27835a920a61784c9c32f8                             0.1s
 => => extracting sha256:6912e23eb44595b44dc6d9882b34f305ad6f6e9667cec70b13e5dbc4f8cd97f4                             0.5s
 => => extracting sha256:bbc5a12237474ec0b23b5051260d56d6436a24d70d9cad961adebe7a037ed813                             0.0s
 => [internal] load build context                                                                                     0.1s
 => => transferring context: 416B                                                                                     0.0s
 => [2/5] WORKDIR /app                                                                                                0.1s
 => [3/5] COPY requirements.txt .                                                                                     0.0s
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                         18.9s
 => [5/5] COPY app.py .                                                                                               0.1s
 => exporting to image                                                                                                3.6s
 => => exporting layers                                                                                               1.4s
 => => exporting manifest sha256:608cfa5aa786bba47116ca08d1570ee8b947500355d86c045b4118452500ba2c                     0.0s
 => => exporting config sha256:0c5f3a4a6055eed093136896793415726cbcdeca01de548af11715bf3ec4c1a9                       0.0s
 => => exporting attestation manifest sha256:ac72436545a1013b576d8f33ae83f44fd5384f8580721176ea46c93171790ef2         0.0s
 => => exporting manifest list sha256:f30bdd57a30398a9bef4b2932520944600326c93915ac8e70d545d031f895e77                0.0s
 => => naming to docker.io/library/student-k8s-demo:v1                                                                0.0s
 => => unpacking to docker.io/library/student-k8s-demo:v1                                                             2.1s
```

```
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> docker images
                                                                                    i Info →  U In Use
IMAGE                                          ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/desktop-cloud-provider-kind:v0.7.0      f078b9ccf478       628MB         171MB    U
docker/desktop-containerd-registry-mirror:v0.0.4  1de142695bb0        52.9MB        16.9MB    U
envoyproxy/envoy:v1.36.7                       29d778ba078e       253MB         63.9MB
kindest/node:v1.36.1                           3489c7674813       1.31GB        383MB    U
student-k8s-demo:v1                            f30bdd57a303       216MB         52.4MB
```

Step 9 -Run and test the container

```
1.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo>
df58339fc8100544079fdd6a0e1ae0facdc9b9433600d74254c71d5416427c35
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> docker ps
CONTAINER ID   IMAGE                COMMAND           CREATED          STATUS          PORTS
       NAMES
df58339fc810   student-k8s-demo:v1   "python app.py"   57 seconds ago   Up 56 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->500
0/tcp   student-demo
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> docker logs student-demo
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [14/Oct/2026 13:42:08] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [14/Oct/2026 13:42:08] "GET /favicon.ico HTTP/1.1" 404 -
172.17.0.1 - - [14/Oct/2026 13:42:27] "GET /health HTTP/1.1" 200 -
```

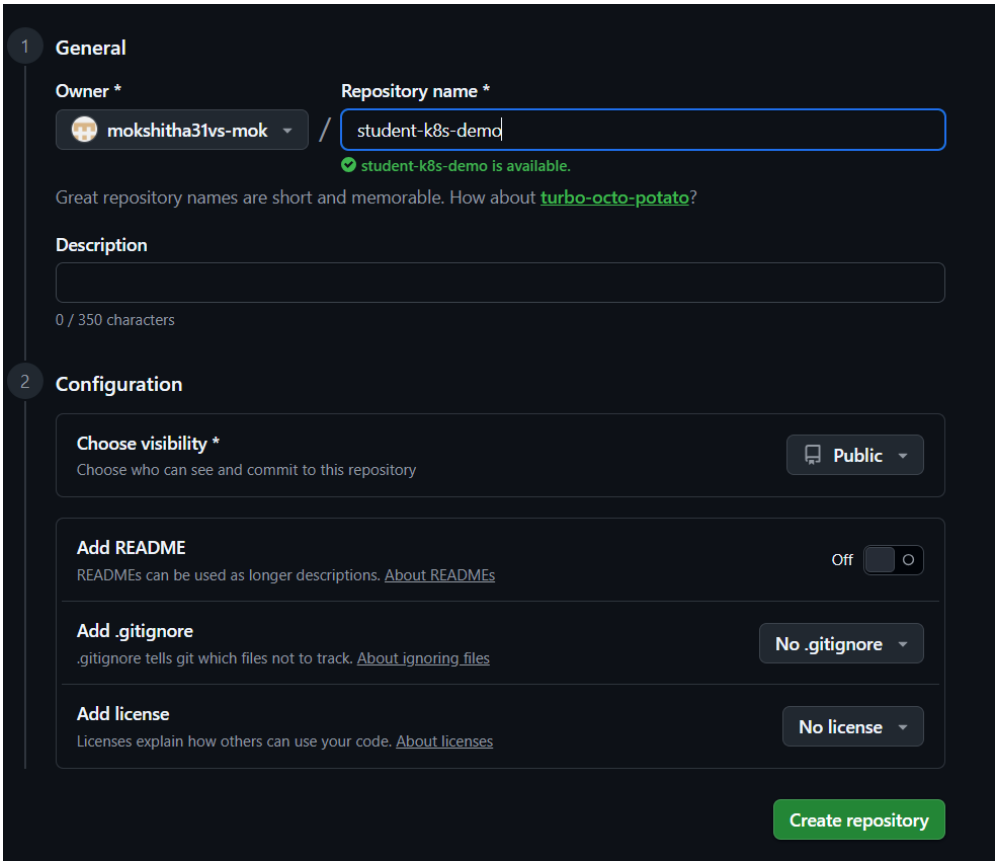




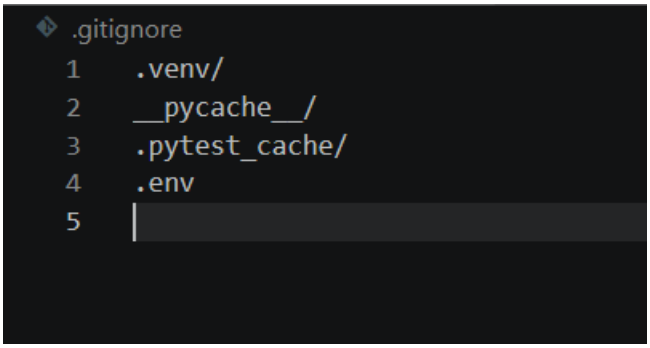
Push source code from VS Code to GitHub:

Step 10 -Create the repository

1. Sign in to GitHub and choose New repository.
2. 2 Name it student-k8s-demo.
3. For this classroom exercise, select Public.
4. Do not initialize the remote repository with a readme,the local folder already contains the project.
5. Create the repository and copy its https url.

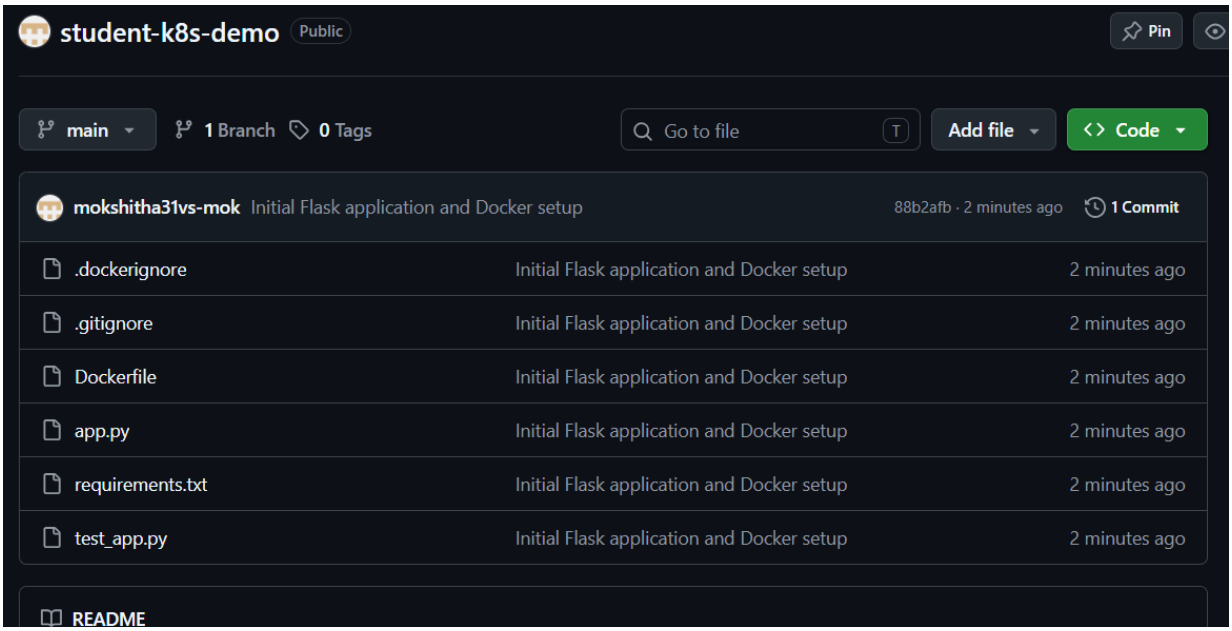


Step 11-Add .gitignore



Push The Project Files To Github:

```
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git init
Initialized empty Git repository in C:/Users/MOKSHITHA/Desktop/student-k8s-demo/.git/
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git add .
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git commit -m "Initial Flask application and Docker setup"
[master (root-commit) 88b2afb] Initial Flask application and Docker setup
 6 files changed, 49 insertions(+)
 create mode 100644 .dockerignore
 create mode 100644 .gitignore
 create mode 100644 Dockerfile
 create mode 100644 app.py
 create mode 100644 requirements.txt
 create mode 100644 test_app.py
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git branch -M main
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git remote add origin https://github.com/mokshitha31vs-mok/student-k8s
-demo.git
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git push -u origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 20 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (8/8), 1.04 KiB | 354.00 KiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mokshitha31vs-mok/student-k8s-demo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```



Add Continuous Integration with GitHub Actions:

Step 12 -Create .github/workflows/pipeline.yml

1. Inside the student-k8s-demo folder create a new folder ".github".
2. Inside the ".github" folder create another new folder "workflows".
3. Inside the "workflows" folder open new file "ci.yml" and add the workflow:

.github > workflows > ! ci.yml

```yaml
name: Flask CI
on:
  push:
    branches: [main]
  pull_request:
    branches: [main]

jobs:
  test:
    runs-on: ubuntu-latest

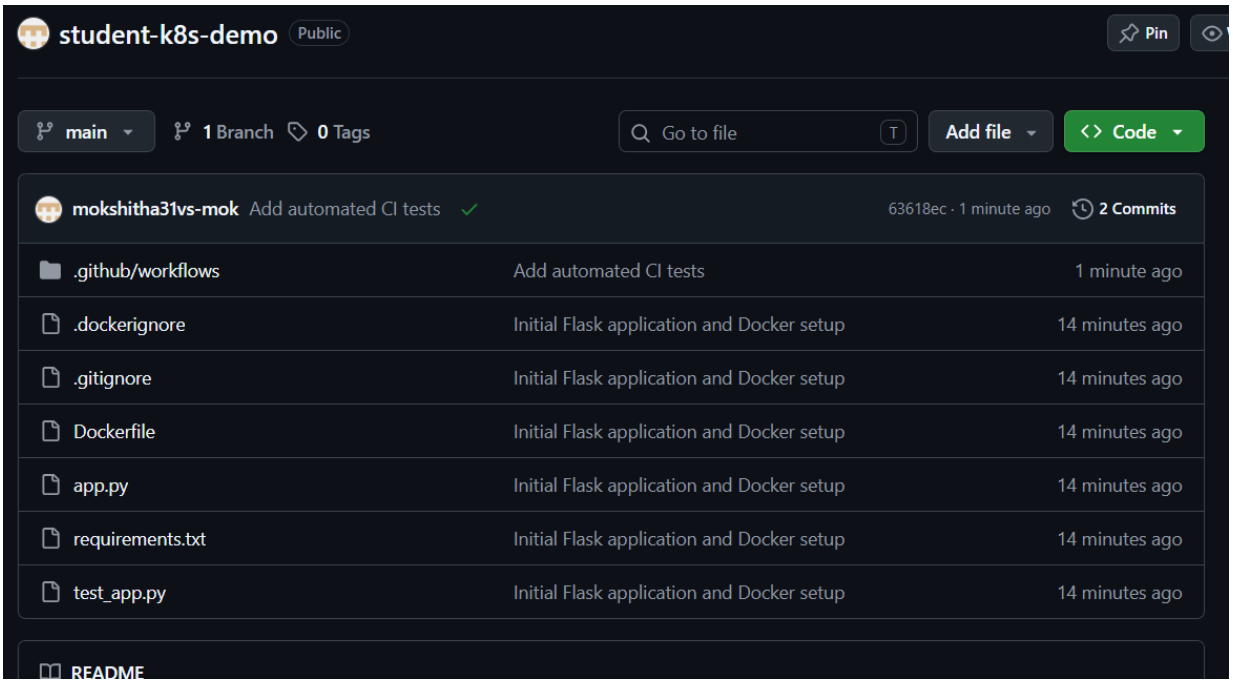
    steps:
      - name: Checkout code
        uses: actions/checkout@v4

      - name: Set up Python
        uses: actions/setup-python@v5
        with:
          python-version: "3.11"

      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install -r requirements.txt
      - name: Run tests
        run: python -m pytest -v
```

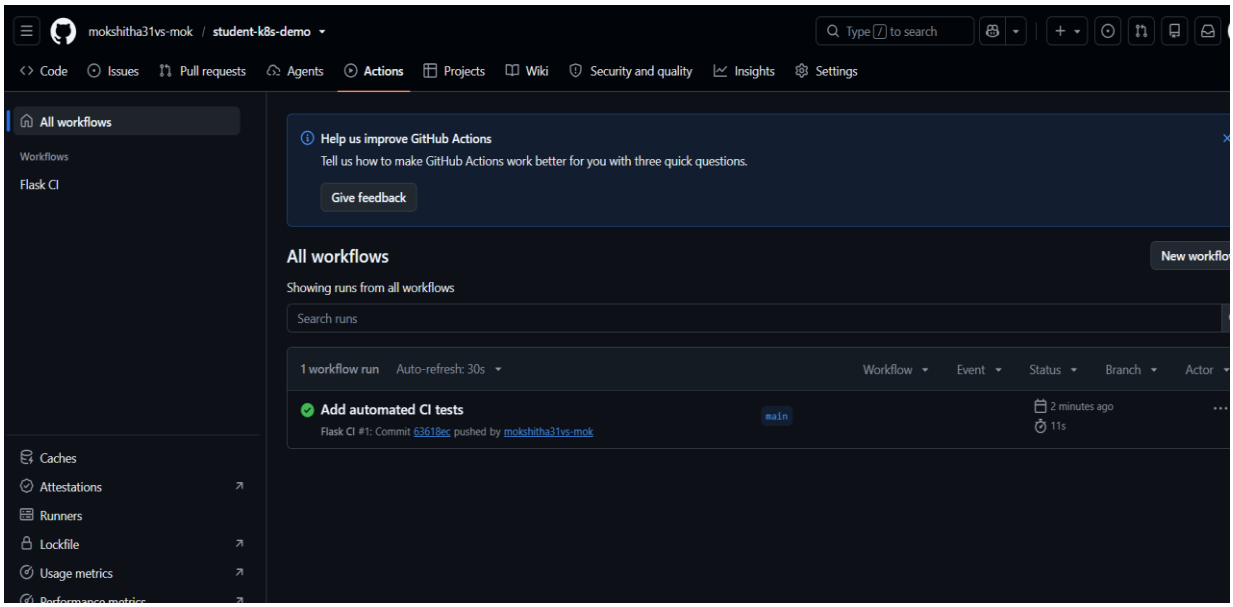
Step 13 - Commit and push the workflow

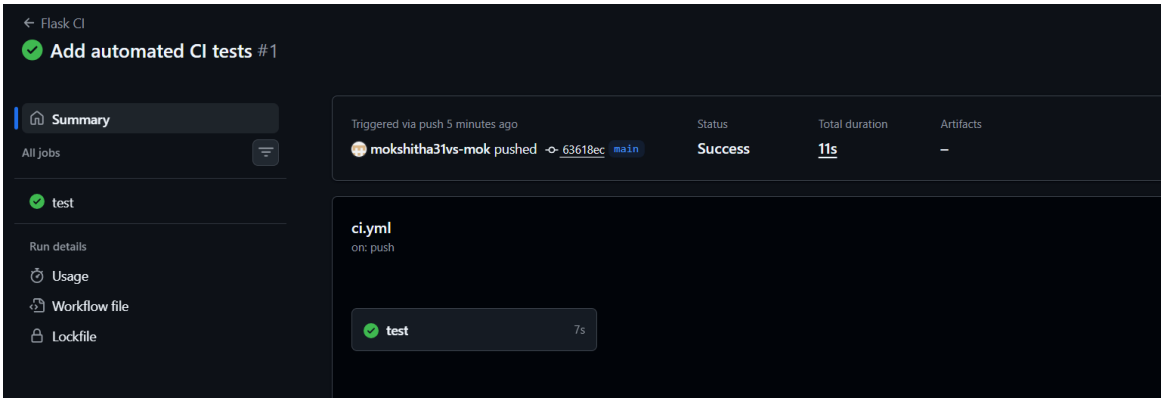
```
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git add .
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git commit -m "Add automated CI tests"
[main 63618ec] Add automated CI tests
 1 file changed, 26 insertions(+)
 create mode 100644 .github/workflows/ci.yml
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git push
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 621 bytes | 621.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mokshitha31vs-mok/student-k8s-demo.git
   88b2afb..63618ec  main -> main
```

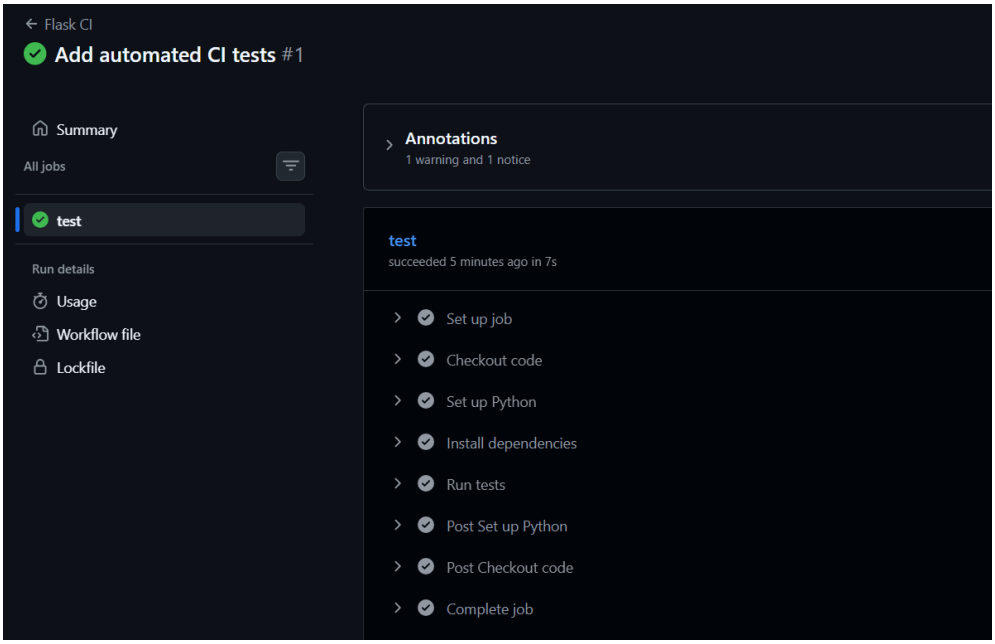


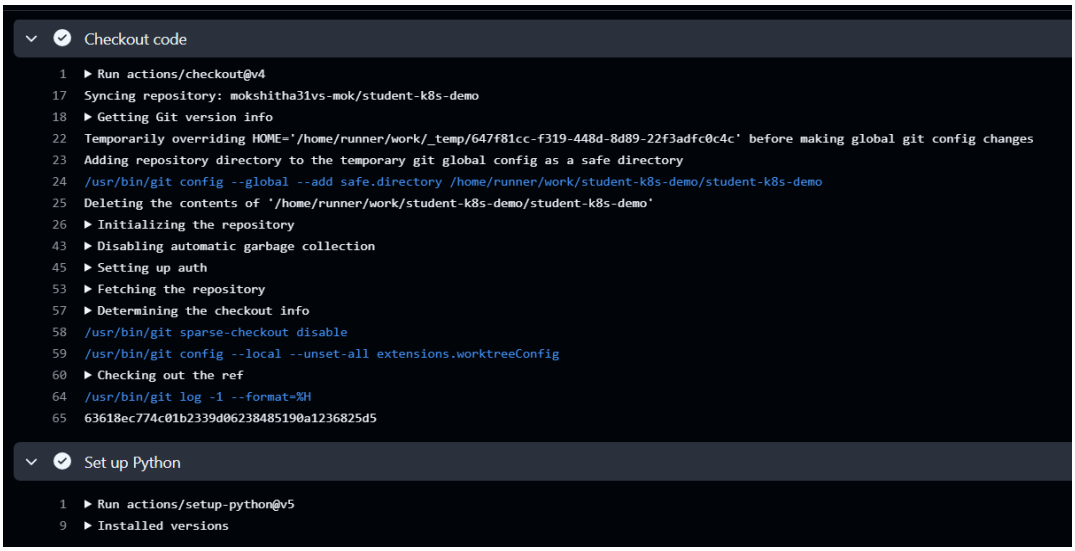
Step 14 -Observe the run :

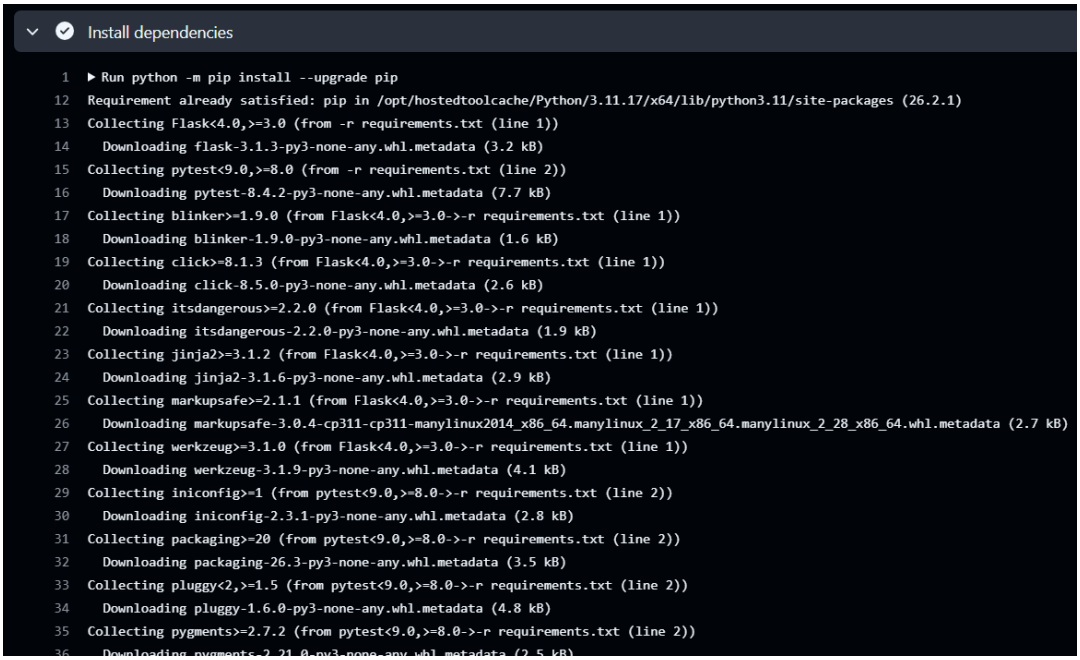
1. Open the repository on GitHub.
2. Click Actions.
3. Open the Student Kubernetes CI/CD workflow run.
4. Expand each step and observe checkout, Python setup, dependency installation, and tests.

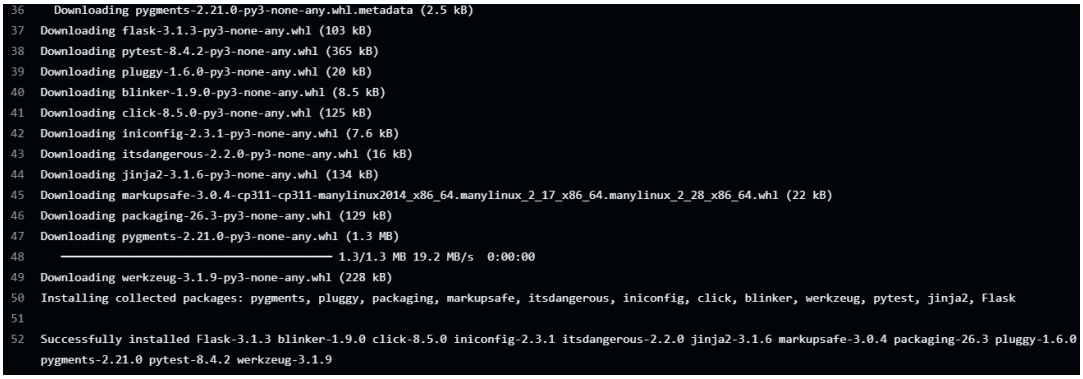

mokshitha31vs-mok / student-k8s-demo
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
All workflows
Workflows
Flask CI
Help us improve GitHub Actions
Tell us how to make GitHub Actions work better for you with three quick questions.
Give feedback
All workflows
New workflo
Showing runs from all workflows
Search runs
1 workflow run
Auto-refresh: 30s
Workflow
Event
Status
Branch
Actor
Add automated CI tests
Flask CI #1: Commit 63618ec pushed by mokshitha31vs-mok
main
2 minutes ago
11s
Caches
Attestations
Runners
Lockfile
Usage metrics


Flask CI
Add automated CI tests #1
Summary
All jobs
test
Run details
Usage
Workflow file
Lockfile
Triggered via push 5 minutes ago
mokshitha31vs-mok pushed 63618ec main
Status
Success
Total duration
11s
Artifacts
–
ci.yml
on: push
test
7s


Flask CI
Add automated CI tests #1
Summary
All jobs
test
Run details
Usage
Workflow file
Lockfile
Annotations
1 warning and 1 notice
test
succeeded 5 minutes ago in 7s
Set up job
Checkout code
Set up Python
Install dependencies
Run tests
Post Set up Python
Post Checkout code
Complete job







Step 15 -Demonstrate a failed test

Temporarily change the expected text in test_app.py to an incorrect value:

```python
test_app.py > test_health_endpoint
from app import app

def test_home_page():
    client = app.test_client()
    response = client.get("/")
    assert response.status_code == 200
    assert b"This message is incorrect" in response.data

def test_health_endpoint():
    client = app.test_client()
    response = client.get("/health")
    assert response.status_code == 200
    assert response.json["status"] == "healthy"
```

Run the test in vs code terminal and push the changed file to github:

```
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> python -m pytest -v
============================================= test session starts =============================================
platform win32 -- Python 3.11.4, pytest-8.4.2, pluggy-1.6.0 -- C:\Users\MOKSHITHA\Desktop\student-k8s-demo\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\MOKSHITHA\Desktop\student-k8s-demo
collected 2 items

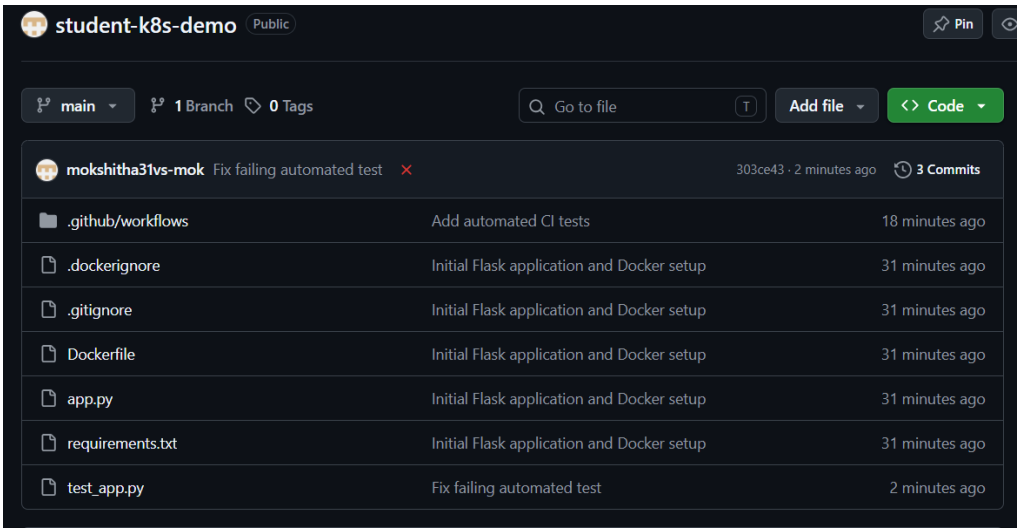
test_app.py::test_home_page FAILED                                                                   [ 50%]
test_app.py::test_health_endpoint PASSED                                                             [100%]

================================================== FAILURES ===================================================
_____________________________________________ test_home_page _____________________________________________

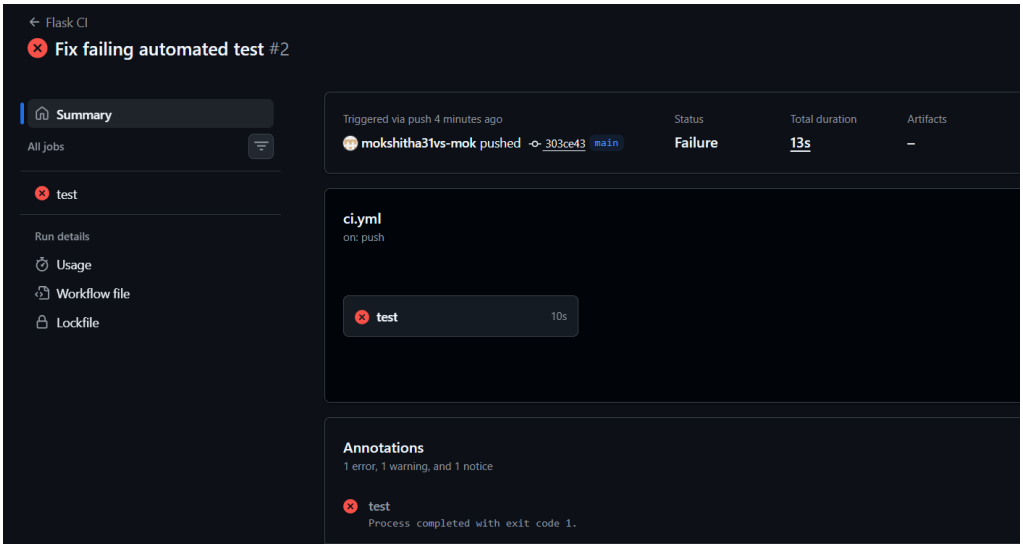
    def test_home_page():
        client = app.test_client()
        response = client.get("/")
        assert response.status_code == 200
>       assert b"This message is incorrect" in response.data
E       AssertionError: assert b'This message is incorrect' in b'Hello Students! Kubernetes CI/CD is working!'
E        +  where b'Hello Students! Kubernetes CI/CD is working!' = <WrapperTestResponse 44 bytes [200 OK]>.data

test_app.py:7: AssertionError
=========================================== short test summary info ===========================================
FAILED test_app.py::test_home_page - AssertionError: assert b'This message is incorrect' in b'Hello Students! Kubernetes CI/CD is working!'
========================================= 1 failed, 1 passed in 0.56s =========================================
```

```
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git add test_app.py
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git commit -m "Fix failing automated test"
[main 303ce43] Fix failing automated test
 1 file changed, 2 insertions(+), 2 deletions(-)
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 345 bytes | 345.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/mokshitha31vs-mok/student-k8s-demo.git
   63618ec..303ce43  main -> main
```



Check the actions in github repository:

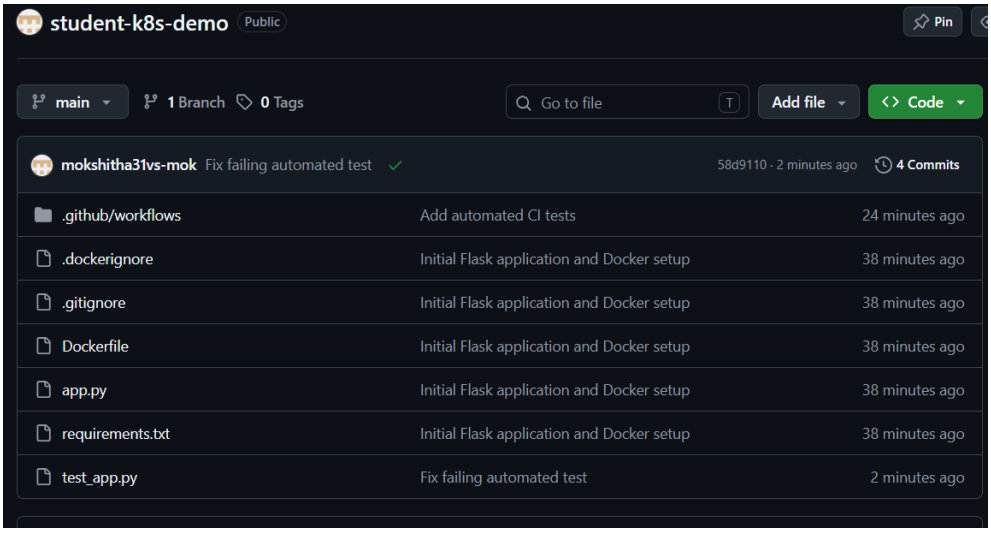


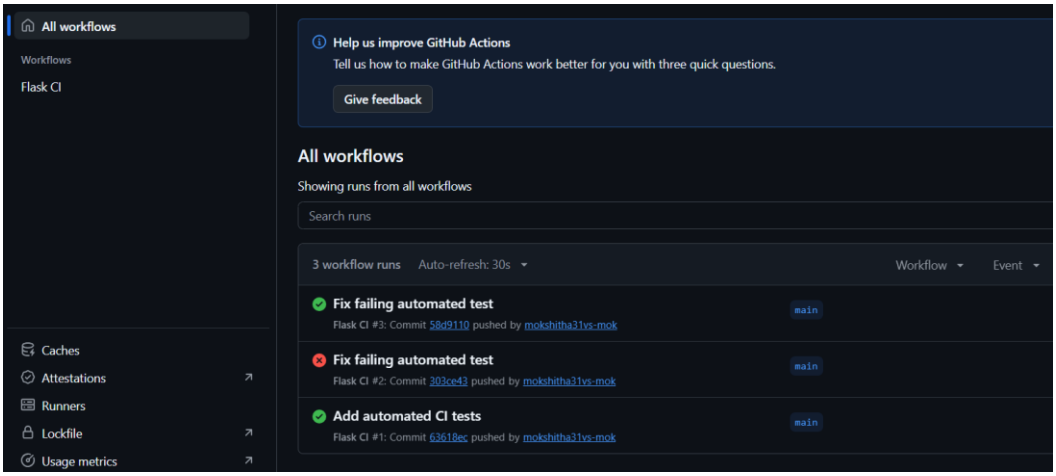
Fix the failed automation by correcting the test_app.py,do test in vs code terminal and push the updated file to github and check actions in github repository.

```
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> python -m pytest -v
============================================== test session starts ==============================================
platform win32 -- Python 3.11.4, pytest-8.4.2, pluggy-1.6.0 -- C:\Users\MOKSHITHA\Desktop\student-k8s-demo\.venv\Scripts\pytho
n.exe
cachedir: .pytest_cache
rootdir: C:\Users\MOKSHITHA\Desktop\student-k8s-demo
collected 2 items

test_app.py::test_home_page PASSED                                                                      [ 50%]
test_app.py::test_health_endpoint PASSED                                                                [100%]

=============================================== 2 passed in 0.24s ===============================================
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git add test_app.py
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git commit -m "Fix failing automated test"
[main 58d9110] Fix failing automated test
 1 file changed, 3 insertions(+), 2 deletions(-)
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 347 bytes | 173.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/mokshitha31vs-mok/student-k8s-demo.git
   303ce43..58d9110  main -> main
```

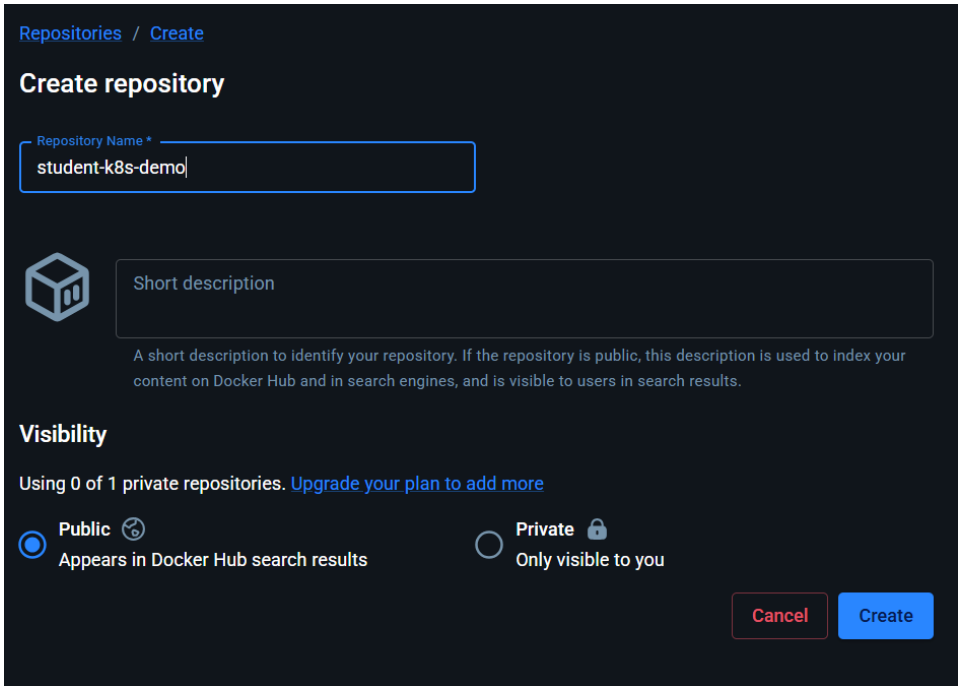




Build and publish the Docker image automatically:

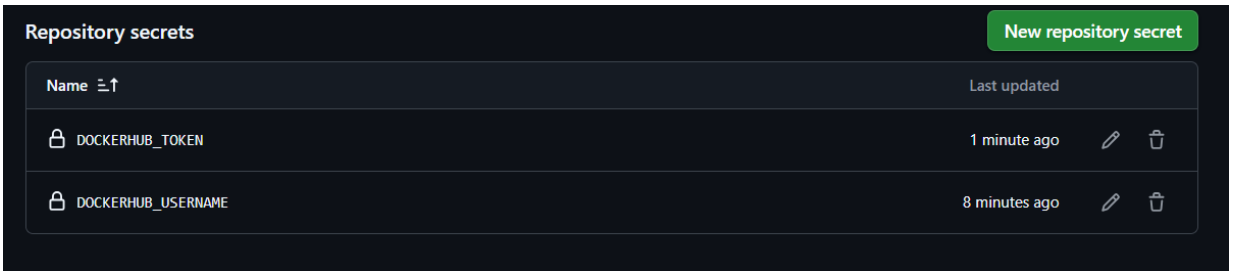
Step 16 -Create a Docker Hub repository

1. Sign in to Docker Hub and create a repository named student-k8s-demo.
2. Make it Public for this introductory lab so the local cluster can pull it without registry credentials.
3. Create an access token with the minimum permissions needed to push the image.



Step 17 -Add GitHub repository secrets

In GitHub, open Settings → Secrets and variables → Actions → New repository secret. Add:



Step 18 -Replace pipeline.yml with this workflow

.github > workflows > ! ci.yml

```yaml
name: Student Kubernetes CI/CD
on:
  push:
    branches: [main]
  pull_request:
    branches: [main]

permissions:
  contents: read

jobs:
  test:
    name: Checkout and Test
    runs-on: ubuntu-latest
    steps:
      - name: Checkout source code
        uses: actions/checkout@v4

      - name: Set up Python
        uses: actions/setup-python@v5
        with:
          python-version: "3.12"
```

```yaml
      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install -r requirements.txt

      - name: Run tests
        run: python -m pytest -v

  docker:
    name: Build and Push Docker Image
    needs: test
    if: github.event_name == 'push'
    runs-on: ubuntu-latest
    steps:
      - name: Checkout source code
        uses: actions/checkout@v4

      - name: Set up Docker Buildx
        uses: docker/setup-buildx-action@v3

      - name: Login to Docker Hub
        uses: docker/login-action@v3
        with:
          username: ${{ secrets.DOCKERHUB_USERNAME }}
          password: ${{ secrets.DOCKERHUB_TOKEN }}

```

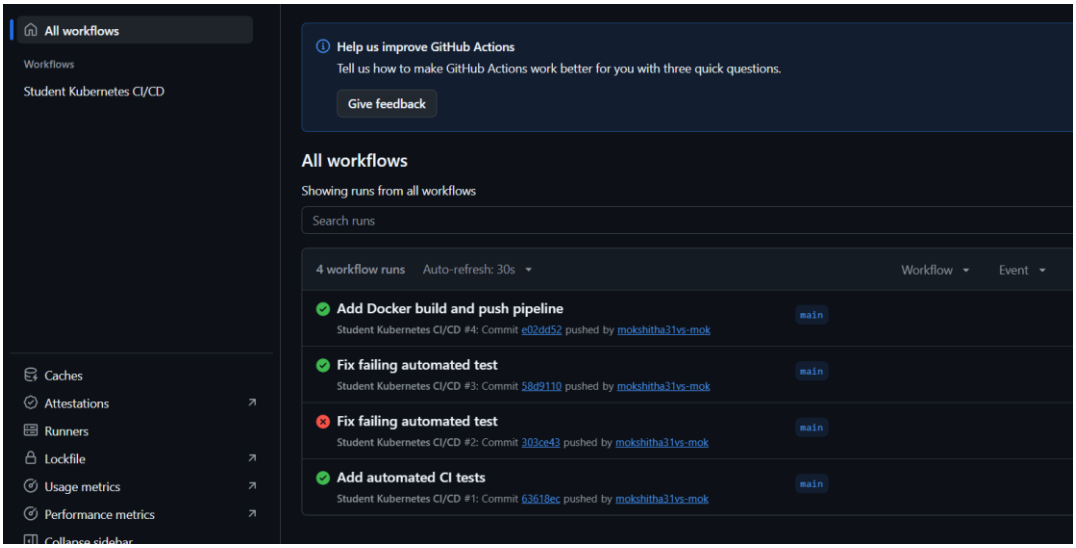
```yaml
      - name: Build and push image
        uses: docker/build-push-action@v6
        with:
          context: .
          push: true
          tags: ${{ secrets.DOCKERHUB_USERNAME }}/student-k8s-demo:${{ github.sha }}

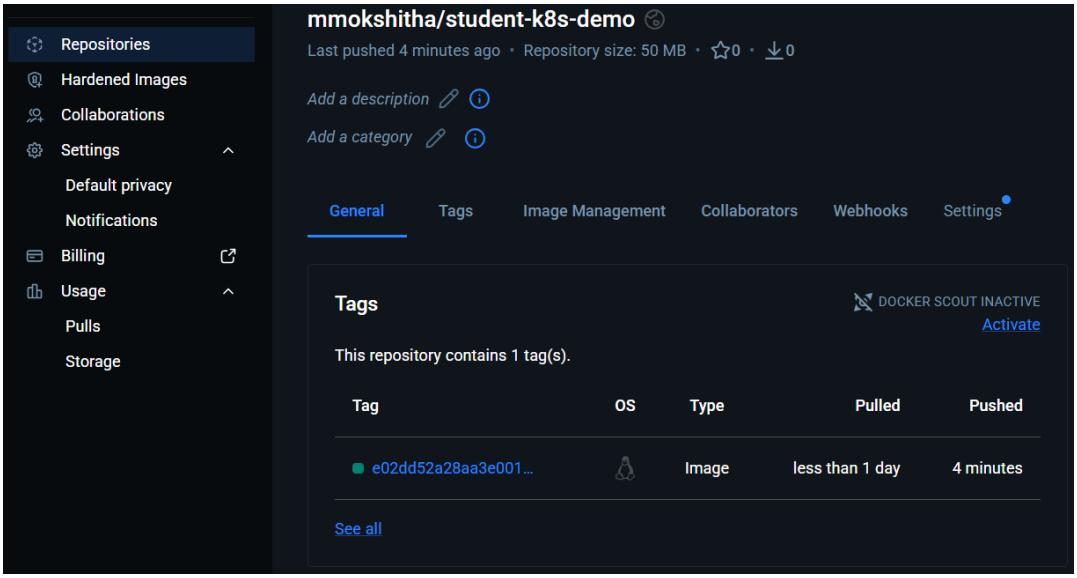
```

Push the Replaced workflow to github:

```
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git add .github/workflows/ci.yml
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git commit -m "Add Docker build and push pipeline"
[main e02dd52] Add Docker build and push pipeline
 1 file changed, 33 insertions(+), 4 deletions(-)
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 877 bytes | 219.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mokshitha31vs-mok/student-k8s-demo.git
   58d9110..e02dd52  main -> main
```

Check Actions:





Deploy the image to Kubernetes:

Step 19 -Create k8s/deployment.yaml

Create a new folder “k8s” inside the “student-k8s-demo” folder and open a new file “deployment.yaml” inside the “k8s” folder.

k8s > ! deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: student-k8s-demo
spec:
  replicas: 2
  selector:
    matchLabels:
      app: student-k8s-demo
  template:
    metadata:
      labels:
        app: student-k8s-demo
    spec:
      containers:
        - name: student-k8s-demo
          image: mmokshitha/student-k8s-demo:e02dd52a28aa3e001bee2f08128aabacf346bfc9
          ports:
            - containerPort: 5000
          readinessProbe:
            httpGet:
              path: /health
              port: 5000
            initialDelaySeconds: 5
            periodSeconds: 5

```

Step 20 - Create k8s/service.yaml

Open a new file “service.yaml” inside “k8s” folder

k8s > ! service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: student-k8s-demo
spec:
  type: NodePort
  selector:
    app: student-k8s-demo
  ports:
    - port: 80
      targetPort: 5000
      nodePort: 30080

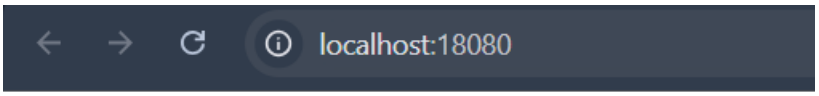
```

Step 21 - Apply the resources

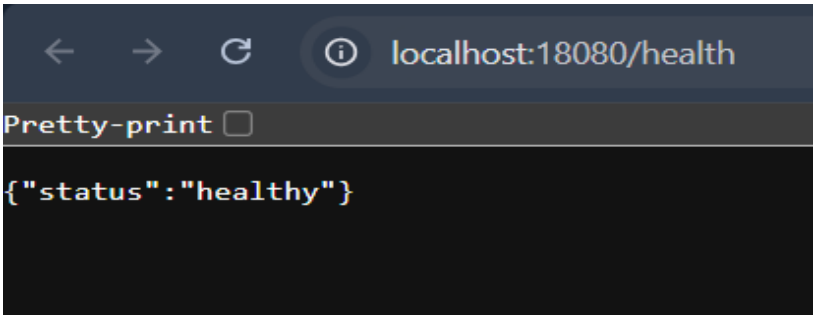
```
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> kubectl config current-context
docker-desktop
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> kubectl get nodes
NAME                 STATUS   ROLES           AGE    VERSION
desktop-control-plane   Ready    control-plane   3h6m   v1.36.1
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> kubectl apply -f k8s/deployment.yaml
deployment.apps/student-k8s-demo created
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> kubectl apply -f k8s/service.yaml
service/student-k8s-demo created
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> kubectl rollout status deployment/student-k8s-demo
deployment "student-k8s-demo" successfully rolled out
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> kubectl get deployments
NAME              READY   UP-TO-DATE   AVAILABLE   AGE
student-k8s-demo   2/2     2            2           70s
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
student-k8s-demo-6cfcdf5f8d-6zz94   1/1     Running   0          82s
student-k8s-demo-6cfcdf5f8d-hl64g   1/1     Running   0          82s
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> kubectl get services
NAME              TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
kubernetes        ClusterIP   10.96.0.1      <none>        443/TCP        3h8m
student-k8s-demo   NodePort    10.96.244.50   <none>        80:30080/TCP   62s
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo>
```

Step 22 -Test the application

```
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> kubectl port-forward service/student-k8s-demo 18080:80
Forwarding from 127.0.0.1:18080 -> 5000
Forwarding from [::1]:18080 -> 5000
Handling connection for 18080
Handling connection for 18080
Handling connection for 18080
Handling connection for 18080
```



Hello Students! Kubernetes CI/CD is working!



Demonstrate an application update:

Step 23 -Change the application and save the app.py file

app.py 1, M | 127.0.0.1:5000 | .gitignore | ci.yml

app.py > home

```python
from flask import Flask, jsonify

app = Flask(__name__)

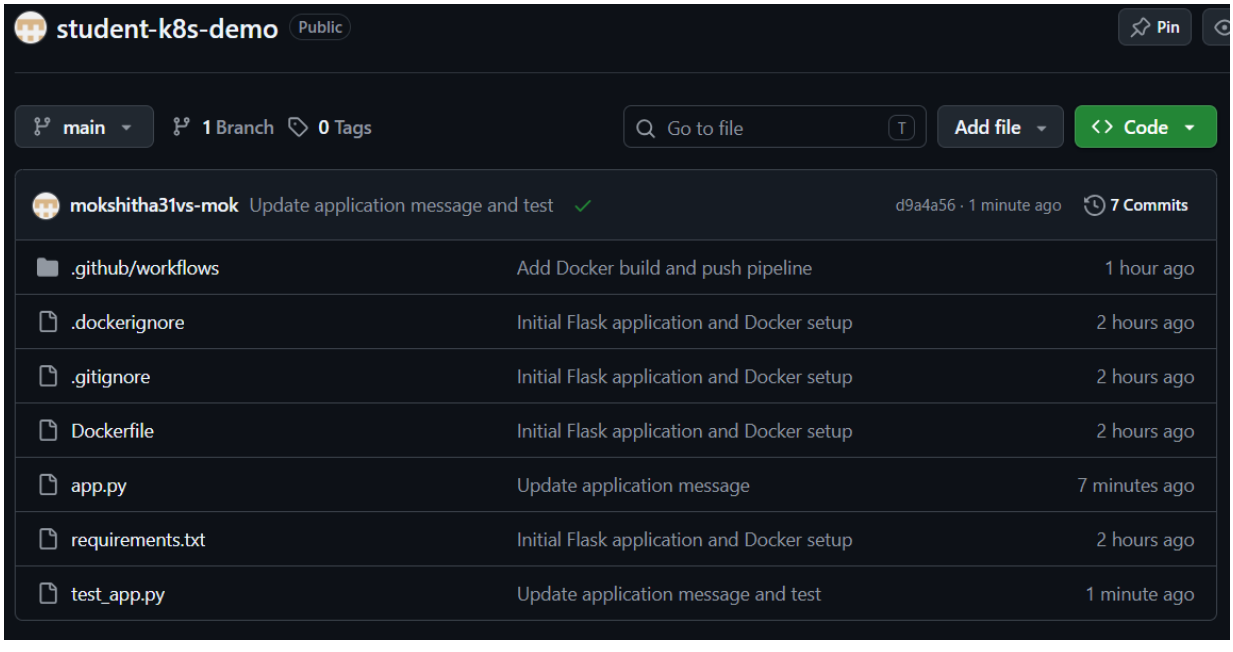
@app.route("/")
def home():
    return "Version 2: Updated through CI/CD!"

@app.route("/health")
def health():
    return jsonify(status="healthy"), 200

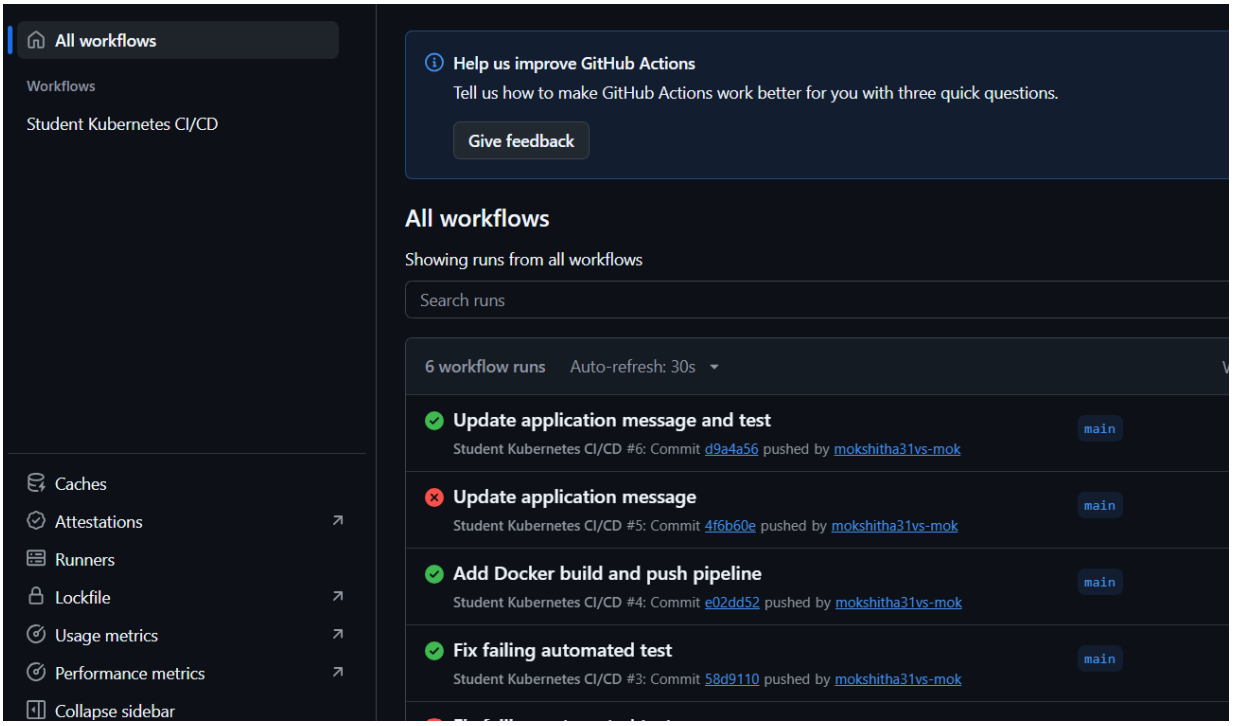
if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

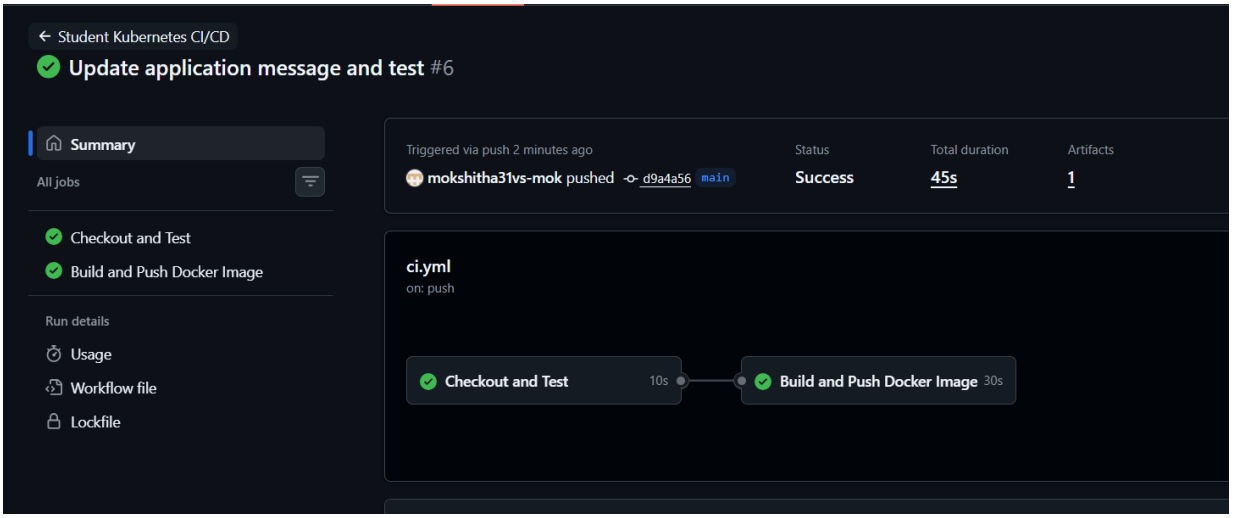
Push the updated file to github:

```
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git add app.py
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git commit -m "Update application message"
[main 4f6b60e] Update application message
 1 file changed, 1 insertion(+), 1 deletion(-)
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 331 bytes | 331.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/mokshitha31vs-mok/student-k8s-demo.git
   e02dd52..4f6b60e  main -> main
```



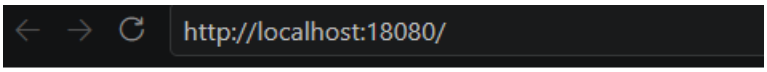
Check Actions:





After GitHub Actions passes, copy the new commit SHA and update the Deployment to use the new image tag:

```
(.venv) PS C:\Users\MOKSHITHA\Desktop\student-k8s-demo> kubectl port-forward pod/student-k8s-demo-7578964d68-lgswl 18080:5000
Forwarding from 127.0.0.1:18080 -> 5000
Forwarding from [::1]:18080 -> 5000
Handling connection for 18080
Handling connection for 18080
```



Version 2: Updated through CI/CD!

